Operations Research for Health Care

Lesson 11/15

João Flávio de Freitas Almeida

LEPOINT: Laboratório de Estudos em Planejamento de Operações Integradas
Departmento de Engenharia de Produção

Universidade Federal de Minas Gerais
Escola de Engenharia

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Outline of this lecture

Facility Location Under Uncertainty

Robust Optimization

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Robust Optimization

Uncertainty regarding the location of facilities

Facilities typically operate for years or decades. Costs, requirements and travel times *can change* significantly [1]. Decisions about the location of facilities are costly and difficult to reverse, and their effects extend over a long time horizon [Snyder, 2006].

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Parameter estimates may be inaccurate due to poor measurements or aggregation of demand points and choice of distance norm. We explore the two-stage nature of facility location problems (choosing locations now and assigning patients to facilities once uncertainty has been resolved).

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The strategic phase involves capital investment, followed by a tactical phase, with uncertainty being resolved in between. But firstly, the terms **certainty**, **risk** and **uncertainty** must be defined [Contreras and Ortiz-Astorquiza, 2019].

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Parameter-based decision type definitions

Certainty parameters are deterministic and known;

[2] The existence of a probabilistic description of uncertainty does not prevent the use of robustness measures.

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Risk Uncertain parameters with known probability distributions. Stochastic programming optimises the *expected* objective function. Parameters are probability distributions. *Identifying scenarios is difficult* (small numbers of scenarios for computations), but the advantage is that parameters are statistically dependent. E.g. demand vs time or locations; costs vs suppliers;

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Uncertainty Parameters are uncertain and probabilities are unavailable [2]. This is often solved with robust optimisation, which optimises worst-case performance. If no probability information is known, *invervals* are assumed.

[2] The existence of a probabilistic description of uncertainty does not prevent the use of robustness measures.

Attitude towards risk and Decisions

The decision maker's **attitude to risk** are:

- Risk neutral: the decision maker does not consider risk when making a decision, and a linear function represents the associated return. If a probability can be assigned to each scenario, a risk-neutral decision maker will seek a decision that *minimises the expected cost (or maximises the expected return)*.
- Risk averse: The decision maker wants to avoid unnecessary risk. The expected value of future assets is no longer an appropriate objective. Such a decision maker may seek to *minimise the maximum cost across all scenarios*.

[3] Location decisions, which are strategic.

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The **decisions** are:

- *ex ante*[3] (here and now): decisions that need to be taken before uncertainty is revealed;
- *ex post*[4] (wait and see): decisions to be taken after the uncertainty has been revealed.

[3] Location decisions, which are strategic.
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Robust optimization

- Choosing an appropriate *performance measure* is part of the modelling process. (e.g. Min E(cost), from P-median, are relatively easy to solve, but Minmax, from P-centre, are difficult).
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- Robust location problems: Discrete parameters: scenario approach. Continuous parameters: lie in some pre-specified interval.
  Minmax cost: minimises the maximum cost over all scenarios (too conservative). Poor solutions for small, medium, large demand.
  Minmax regret (opportunity loss): the difference (absolute or percentage) between the cost of solving a scenario and the cost of the optimal solution for that scenario.

Model and solution robustness [Mulvey et al., 1995]

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To illustrate, we work with some well-known problems. For a comprehensive overview of the foundations of robust optimization, please refer to the seminal works of [Kouvelis et al., 1997] and [Ben-Tal, 2009], and lecture notes of [Oliveira, 2024].

P-median problem [Contreras and Ortiz-Astorquiza, 2019]

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$$\text{min.} \sum_{i}^{I} \sum_{j}^{J} D_i A_{ij} x_{ij} \tag{8.1}$$

$$\text{s.t.:} \sum_{j} x_{ij} = 1 \quad \forall\, i \in I \tag{8.2}$$

$$x_{ij} \leq x_{jj} \quad \forall\, i \in I, j \in J \tag{8.3}$$

$$\sum_{j} x_{jj} = P \tag{8.4}$$

$$x_{ij} \in \{0,1\} \quad \forall\, i \in I, j \in J \tag{8.5}$$

The Minmax P-median[5] [Contreras and Ortiz-Astorquiza, 2019]

I : The set of all patients' origins.
J : The set of all (possible) destination health care locations.
W : Finite set of scenarios ω in Ω.

[5] The Robust Optimization Minmax Cost and Minmax Regret P-median problems work with a finite set of scenarios.

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- $y_j \in \{0, 1\}$: Decision to locate the health care facility in $j \in J$ (or not).
- v: Auxiliary variable for robust optimisation.

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The Minmax **Cost** P-median[6] [Contreras and Ortiz-Astorquiza, 2019]

$$\text{min. } v \tag{8.6}$$

$$\text{s.t.: } \sum_{i}^{I} \sum_{j}^{J} D_{i\omega} A_{ij\omega} x_{ij\omega} \leq v \quad \forall\, \omega \in \Omega \tag{8.7}$$

$$\sum_{j} x_{ij\omega} = 1 \quad \forall\, i \in I, \omega \in \Omega \tag{8.8}$$

$$x_{ij\omega} \leq y_j \quad \forall\, i \in I, j \in J, \omega \in \Omega \tag{8.9}$$

$$\sum_{j} y_j = P \tag{8.10}$$

$$x_{ij\omega} \in \{0, 1\} \quad \forall\, i \in I, j \in J, \omega \in \Omega \tag{8.11}$$

$$y_j \in \{0, 1\} \quad \forall j \in J \tag{8.12}$$

[6] Solution overly conservative. Complete risk aversion across all scenarios.

The Minmax **Regret** P-median[7] [Contreras and Ortiz-Astorquiza, 2019]

$$\text{min. } v \tag{8.6}$$

$$\text{s.t.: } \sum_{i}^{I}\sum_{j}^{J} D_{i\omega}A_{ij\omega}x_{ij\omega} - v^*_\omega \leq v \quad \forall\, \omega \in \Omega \tag{\textbf{8.13}}$$

$$\text{or s.t.: } \frac{\sum_i^I \sum_j^J D_{i\omega}A_{ij\omega}x_{ij\omega} - v^*_\omega}{v^*_\omega} \leq v \quad \forall\, \omega \in \Omega \tag{\textbf{8.14}}$$

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[7] Minimizes the maximum regret of all scenarios. v^*_ω is the solution of (8.1)–(8.5) for each $\omega \in \Omega$.

The Uncapacitated Facility Location Problem (UFLP)[Contreras and Ortiz-Astorquiza, 2019]

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- F_j: Setup cost for facility $j \in J$ (\$).
- C_{ij}: Unit cost for meeting demand (not full demand) from $i \in I$ to $j \in J$ (\$).

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- $x_{ij} \geq 0$: The fraction of demand of $i \in I$ supplied from facility $j \in J$.
- $y_j \in \{0,1\}$: Decision to locate the health care facility in $j \in J$ (or not).

$$\text{min. } \sum_{j}^{J} F_j y_j + \sum_{i}^{I} \sum_{j}^{J} C_{ij} \bar{D}_i x_{ij} \tag{8.15}$$

$$\text{s.t.: } \sum_{j} x_{ij} = 1 \quad \forall\, i \in I \tag{8.16}$$

$$x_{ij} \leq y_j \quad \forall\, i \in I, j \in J \tag{8.17}$$

$$y_j \in \{0,1\} \quad \forall\, j \in J \tag{8.18}$$

$$x_{ij} \geq 0 \quad \forall\, i \in I, j \in J \tag{8.19}$$

The Robust UFLP[9] with **Box** or **Ellipsoidal** uncertainty set

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I : The set of all patients' origins.
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[8] $L = 1$ is the largest ellipsoid contained in $\mathcal{U}^B$. $L = \sqrt{|I|}$ is the smallest ellipsoid containing $\mathcal{U}^B$

[9] A way of capturing infinite uncertainty set in these cases is via intervals (Box set or Ellipsoid set).

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- $\bar{D}_i$: Population *reference value* requirement for health at origin $i \in I$ (pop).
- ϵ: Uncertainty magnitude. Multiplied by a scaling factor Δ_i, often equals D_i(%)
- L: A maximum level imposed for the total scaled variation[8].
- D_i: Uncertain health requirement at $i \in I$. Lies in a interval $\mathcal{U}_i^B$ or $\mathcal{U}_i^E$ (pop).
- $\mathcal{U}^B = \{D_i \in \mathbb{R} | -1 \leq \frac{D_i - \bar{D}_i}{\epsilon \bar{D}_i} \leq 1 \quad \forall\, i \in I\}$: Uncertainty Box set.
- $\mathcal{U}^E = \{D_i \in \mathbb{R} | \sum_{i \in I} \left[\frac{D_i - \bar{D}_i}{\epsilon \bar{D}_i}\right] \leq L^2\}$: Uncertainty Ellipsoid set.

[8] $L = 1$ is the largest ellipsoid contained in $\mathcal{U}^B$.$L = \sqrt{|I|}$ is the smallest ellipsoid containing $\mathcal{U}^B$

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The Robust UFLP with **Box** or **Ellipsoidal** uncertainty set

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The UFLP[10] with **Box** or **Ellipsoidal** uncertainty set

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$$\text{min. } v \tag{8.20}$$

$$\text{s.t.: } \sum_{j}^{J} F_j y_j + \sum_{i|D_i \in \mathcal{U}^B}^{I} \sum_{j}^{J} C_{ij}[D_i(1+\epsilon)]x_{ij} \leq v \tag{\textbf{8.23}}$$

$$\text{or s.t.: } \sum_{j}^{J} F_j y_j + \sum_{i|D_i \in \mathcal{U}^E}^{I} \sum_{j}^{J} C_{ij}[D_i(1+\epsilon)]x_{ij} \leq v \tag{\textbf{8.23}}$$

$$\text{s.t.: } \sum_{j} x_{ij} = 1 \quad \forall\, i \in I \tag{8.16}$$

$$x_{ij} \leq y_j \quad \forall\, i \in I, j \in J \tag{8.17}$$

$$y_j \in \{0,1\} \quad \forall\, j \in J \tag{8.18}$$

$$x_{ij} \geq 0 \quad \forall\, i \in I, j \in J \tag{8.19}$$

[10] Under box or ellipsoidal uncertainty set, the constraint (8.23) should replace $\mathcal{U}^B$ with $\mathcal{U}^E$.

The α-Robustness[11] for P-median [Contreras and Ortiz-Astorquiza, 2019]

I : The set of all patients' origins.
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[11] When probabilities can be associated with the scenarios.

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The α-Robustness[12] for P-median[13] [Contreras and Ortiz-Astorquiza, 2019]

$$\text{min.} \sum_{\omega}^{\Omega}\sum_{i}^{I}\sum_{j}^{J}\pi_w D_{i\omega}A_{ij\omega}x_{ij\omega} \tag{8.25}$$

$$\text{s.t.:} \sum_{i}^{I}\sum_{j}^{J} D_{i\omega}A_{ij\omega}x_{ij\omega} \leq (1+\alpha)v_\omega^* \quad \forall\, \omega \in \Omega \tag{8.26}$$

$$\sum_{j} x_{ij\omega} = 1 \quad \forall\, i \in I, \omega \in \Omega \tag{8.8}$$

$$x_{ij\omega} \leq y_j \quad \forall\, i \in I, j \in J, \omega \in \Omega \tag{8.9}$$

$$\sum_{j} y_j = P \tag{8.10}$$

$$x_{ij\omega} \in \{0,1\} \quad \forall\, i \in I, j \in J, \omega \in \Omega \tag{8.11}$$

$$y_j \in \{0,1\} \quad \forall j \in J \tag{8.12}$$

[12] The relative regret should be $\leq \alpha$

[13] The v_ω^* is the solution of (8.6)–(8.12) for each $\omega \in \Omega$.

References

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